SSC MTS 2024 question paper with answers

8 October 2024 · Shift 3

79 questions as the Staff Selection Commission (SSC) printed them · answers from the official key · marking: 3 marks a question; no negative marking in Session-I (Numerical and Mathematical Ability, Reasoning), 1 mark deducted for each wrong answer in Session-II (General Awareness, English).

The other 11 of the paper's 90 questions are not reproduced: they need a figure or are printed as an image, or ask about formatting the source does not keep.

Try every question first: the answer key and the explanations follow the questions. Official answer-key notice: https://ssc.gov.in/api/attachment/uploads/masterData/AnswerKeys/approved_writeup_26425.pdf

Numerical and Mathematical Ability 15 questions

Q.1

The average weight of a family of five members (in kg) whose weights are 40 kg, 49 kg, 56 kg, 65 kg and 38 kg is:

(A) 50.6
(B) 48.6
(C) 51.6
(D) 49.6

Q.2

360 toffees are distributed among Pushkal, Quadir and Rustam in the ratio 2 : 3 : 4. After distribution, Quadir gives 20% of his toffees to Pushkal and Rustam gives 25% of his toffees to Pushkal. What is the new ratio of toffees with Pushkal, Quadir and Rustam?

(A) 6 : 4 : 5
(B) 3 : 5 : 7
(C) 5 : 4 : 6
(D) 7 : 3 : 5

Q.3

Reena sells an article at 12% below its cost price. Had she sold it for ₹540 more, she would have gained 15%. At what price should she sell the article to gain 20%?

(A) ₹2,004
(B) ₹4,200
(C) ₹2,400
(D) ₹2,040

Q.6

A fast train takes 1 hour less than a slow train for a journey of 96 km. If the speed of the slow train is 8 km/h less than that of the fast train, then find the speeds of the fast train and slow train respectively.

(A) 12 km/h, 14 km/h
(B) 32 km/h, 24 km/h
(C) 26 km/h, 34 km/h
(D) 34 km/h, 28 km/h

Q.7

A car travels 80 km at the speed of 20 km/h and the next 30 km at the speed of 30 km/h. What is its average speed?

(A) 40 km/h
(B) 30 km/h
(C) 20 km/h
(D) 22 km/h

Q.8

There are 40 students in a class. Their average weight is 60 kg. Two students left the class. Then the average weight is reduced by 500 g. Find the weight of the two students who left the class (in kg).

(A) 139
(B) 129
(C) 130
(D) 140

Q.10
The average weight of a family of five members (in kg) whose weights are 40 kg, 49 kg, 56 kg, 66 kg and 38 kg is:

(A) 50.8
(B) 49.8
(C) 51.8
(D) 48.8

Q.11
Find the simple interest (in ₹) on ₹4000 as a sum borrowed at 3% per year rate of interest for 6 years.

(A) 660
(B) 720
(C) 680
(D) 760

Q.12
The centroid of a ΔXYZ is G. The area of ΔXYZ is 45 cm^2. The area of ΔGYZ is:

(A) 15 cm^2
(B) 7.5 cm^2
(C) 30 cm^2
(D) 22.5 cm^2

Q.13
A person got an article at 22% discount, if he got no discount, then he had to pay ₹1,760 more. How much did he pay (in ₹) for the article?

(A) 5,120
(B) 6,360
(C) 6,240
(D) 5,640

Q.14
By selling 20 cakes, a shopkeeper earns selling price of 4 cakes. Determine the profit percentage.

(A) 10%
(B) 20%
(C) 25%
(D) 4%

Q.15
R earns ₹55,000 per month and his annual expenditure is ₹4,50,000. What is his percentage of savings per annum?

(A) 36.21%
(B) 38.12%
(C) 31.82%
(D) 32.81%

Q.16
The mean of marks secured by 3 students in grade A of class XI is 83, that of 4 students of grade B is 76 and that of 5 students of grade C is 85. What will be the mean of marks of the students of three grades of Class XI?

(A) 80.5
(B) 81
(C) 80
(D) 81.5

Q.17
What sum (in ₹) will earn a simple interest of ₹240 in 2 years at 16% per year rate of interest?

(A) 750
(B) 375
(C) 1000
(D) 1500

Q.18
The HCF of 513,1134 and 1215 is:

(A) 27
(B) 43
(C) 45
(D) 19

Reasoning Ability and Problem Solving 15 questions

Q.1

Read the given statements and conclusions carefully. Assuming that the information given in the statements is true, even if it appears to be at variance with commonly known facts, decide which of the given conclusions logically follow(s) from the statements. Statements: All pencils are remotes. Some pencils are ACs. Conclusions: [1] Some ACs are remotes. [2] All remotes are Acs.

(A) Both conclusions [1] and [2] follow
(B) Only conclusion [1] follows
(C) Only conclusion [2] follows
(D) Neither conclusion [1] nor [2] follows

Q.2

Which two signs should be interchanged to make the following equation mathematically correct? 15 × 14 – 8 + 12 ÷ 4 = 5

(A) × and –
(B) ÷ and –
(C) × and +
(D) ÷ and +

Q.5

Read the given statements and conclusions carefully. Assuming that the information given in the statements is true, even if it appears to be at variance with commonly known facts, decide which of the given conclusions logically follow(s) from the statements. Statements: All leaves are trees. Some trees are plants. Conclusions: (I) Some leaves are plants. (II) All leaves are trees.

(A) Only conclusion I follows
(B) Both conclusions I and II follow
(C) Only conclusion II follows
(D) Neither conclusion I nor II follows

Q.8

A, B, C, L, M, Y and X are sitting around a circular table facing the centre (but not necessarily in the same order). Only two people sit between B and L when counted from the right of L. X sits to the immediate left of Y. A sits to the immediate right of M. Y is an immediate neighbour of L. Who sits fourth to the right of C?

(A) A
(B) M
(C) L
(D) X

Q.9

What will come in the place of '?' in the following equation, if '+' and '÷' are interchanged and '×' and '–' are interchanged? 48 × 16 – 20 ÷ 21 + 7 = ?

(A) 269
(B) –269
(C) –259
(D) 259

Q.10

In a certain code language, if HOUSE is coded as 67 and GROUND is coded as 83, then how will KITCHEN be coded?

(A) 119
(B) 130
(C) 110
(D) 122

Q.11

'NQRP' is related to 'TWXV' in a certain way based on the English alphabetical order. In the same way, 'FHIG' is related to 'LNOM'. To which of the following is 'LOPN' related following the same logic?

(A) QTUS
(B) QSTR
(C) RSQT
(D) RUVT

Q.12

What should come in place of the question (?) in the following series? 4 ,-27, -58, -89 ,-120 ,?

(A) -131
(B) -151
(C) 151
(D) -141

Q.13

16 is related to 176 following a certain logic. Following the same logic, 21 is related to 231. To which of the following is 33 related, following the same logic? (NOTE : Operations should be performed on the whole numbers, without breaking down the numbers into its constituent digits. E.g. 13 – Operations on 13 such as adding / subtracting / multiplying, etc., to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.)

(A) 362
(B) 364
(C) 363
(D) 361

Q.14

L, M, N, O, P, Q and R are sitting around a circular table, facing the centre (but not necessarily in the same order). Only three people are sitting between R and O when counted from the left of R. Only M is sitting between L and N. N is an immediate neighbour of O. P is sitting third to the right of L. Who is sitting second to the left of O?

(A) P
(B) L
(C) Q
(D) R

Q.16

Select the option in which the numbers share the same relationship as that shared by the given pair of numbers. 48 : 42

(A) 63 : 28
(B) 48 : 24
(C) 69 : 32
(D) 65 : 28

Q.17

What should come in place of the question mark (?) in the given series based on the English alphabetical order? SXG, QVI, OTK, MRM ,?

(A) KPP
(B) KPO
(C) LPP
(D) LPO

Q.18

In a certain code language, ‘COPY’ is coded as ‘2846’ and ‘APES’ is coded as ‘7321’. What is the code for ’P’ in that language?

(A) 3
(B) 2
(C) 1
(D) 4

Q.19

MHPL is related to OJRN in a certain way based on the English alphabetical order. In the same way, PKSO is related to RMUQ. To which of the following is SNVR related following the same logic?

(A) PUTX
(B) UPXT
(C) PUXT
(D) UPTX

Q.20

In a certain code language, 'MASTER' is coded as '917032', and 'STREAK' is coded as '372159'. What is the code for ‘K’ in 'STREAK'?

(A) 7
(B) 3
(C) 9
(D) 5

General Awareness 24 questions

Q.1

'Sange Meel Se Mulaqat' is the documentary of which famous musician?

(A) Hari Prasad Chaurasia
(B) Zakir Hussain
(C) MS Gopalakrishnan
(D) Bismillah Khan

Q.2

Kaal Baisakhi belongs to which state from the following?

(A) Bihar
(B) West Bengal
(C) Rajasthan
(D) Punjab

Q.3

Which of the following statements is NOT true about Plastids?

(A) They have their own DNA.
(B) Stroma is found in chloroplast.
(C) They have their own ribosomes.
(D) Starch, oils and protein granules are stored in chromoplasts.

Q.4

Which of the following is NOT associated with the Iron Age in India?

(A) Smelting
(B) Mehrauli iron pillar
(C) Northern black polished ware
(D) Painted grey ware

Q.5

Which of the following types of emergencies is declared under Article 360 of the Indian Constitution?

(A) Constitutional Emergency
(B) National Emergency
(C) Financial Emergency
(D) State Emergency

Q.6

Which of the following statements is/are correct about the Union Government budget in India? a. The Budget reflects the vision of the government in fulfilling the aspirations of people. b. The Budget is the summary report of the financial performance of the government in power. c. The Budget provides details of expenditures meted out by various Union Government ministries.

(A) All a, b and c
(B) Only a and b
(C) Only c
(D) Only a

Q.7

Who among the following adopted the policy of consolidation in place of expansion by eliminating the rebels?

(A) Raziya
(B) Nasiruddin Mahmud
(C) Ghiyasuddin Balban
(D) Shamsuddin Iltutmish

Q.8

Which material was largely used to build the stupa at Piprahwa in Uttar Pradesh?

(A) Granite Stone
(B) Marble
(C) Sandstone
(D) Bricks

Q.9

Which of the following committees was set up to investigate into the infamous Jallianwala Bagh massacre?

(A) Sadlar Committee
(B) Hunter Committee
(C) Butler Committee
(D) Henry Fowler Committee

Q.10
In the animal kingdom, which of the following phyla includes animals with a segmented body?

(A) Chordata
(B) Arthropoda
(C) Mollusca
(D) Annelida

Q.11
Which edition of the Pro Kabaddi League was started in December 2023?

(A) 10th
(B) 11th
(C) 12th
(D) 8th

Q.12
Which of the following war dances is performed in Ngada festival and is a performed by the tribes of Nagaland?

(A) Jawara
(B) Rengma
(C) Dalkhai
(D) Garadi

Q.13
A _____ happens in chess when a player gives up material during the initiation to seek some kind of compensation.

(A) Gambit
(B) Blockade
(C) Nick
(D) Castling

Q.14
Two masses of 1 kg and 2 kg were dropped from a height of 3.2 metres. Their respective velocities just before touching the ground will be: (Assume acceleration due to gravity g = 10 m/s^2)

(A) 8 m/s for both
(B) 8 m/s and 4 m/s
(C) 3.2 m/s for both
(D) 4 m/s and 8 m/s

Q.15
_____was launched by the Government of India in 2020 to make the country and its citizens independent and self-sufficient in all respects.

(A) Smart Cities Mission
(B) Swachh Bharat Mission
(C) Digital India Mission
(D) Atmanirbhar Bharat Abhiyan

Q.16
'The Namesake', 'Unaccustomed Earth' and 'The Lowland' are novels written by who among the following authors?

(A) Aravind Adiga
(B) Arundhati Roy
(C) Jhumpa Lahiri
(D) Shobha De

Q.17
In June 2023, which country conferred the 'Order of the Nile' award on Narendra Modi?

(A) Singapore
(B) Lebanon
(C) Turkey
(D) Egypt

Q.18
How many components are there in the New India Literacy Program (NILP) scheme launched by the Union Ministry of Education in March 2023?

(A) Nine
(B) Five
(C) Three
(D) Seven

Q.19

From which of the following countries has the Indian Constitution borrowed the concept of the Concurrent List?

(A) Ireland
(B) Australia
(C) Russia
(D) USA

Q.21

According to National Multidimensional Poverty Index 2023 (A Progress Review), what was the proportion of population in multidimensional poverty (Headcount Ratio) in urban India in 2019-2021?

(A) 6.96%
(B) 2.12%
(C) 11.32%
(D) 5.27%

Q.22

When is the National Unity Day celebrated in India?

(A) 14 November
(B) 2 October
(C) 31 October
(D) 26 January

Q.23

The igneous, sedimentary and metamorphic rocks are interconvertible through a continuous process called:

(A) lithification
(B) rock cycle
(C) metamorphism
(D) weathering and erosion

Q.24

Mallika Sarabhai was honoured with the Padma Bhushan in 2010 for her contribution to which of the following fields?

(A) Literature
(B) Dance
(C) Singing
(D) Painting

Q.25

In which of the followings years was the National Population Policy of India released?

(A) 2000
(B) 2015
(C) 1952
(D) 1995

English Language and Comprehension 25 questions

Q.1

Select the grammatically correct sentence.

(A) A teacher from South explained the theory in the detailed manners.
(B) The teacher from an South explained a theory in a detailed manner.
(C) The teacher from the South explained the theory in a detailed manner.
(D) The teacher from a South explained the theory in the detailed manner.

Q.2

Select the most appropriate option to fill in the blank. The CEO came up with a groundbreaking strategy to _____ new markets and capitalise on emerging trends.

(A) break into
(B) break down
(C) break off
(D) break out

Q.3

Select the most appropriate option that can substitute the underlined segment without changing its meaning in the given sentence. Family planning programs play a crucial role in promoting reproductive health and empowering individuals to make informed choices, “contributing to sustainable population growth” and a healthier society.

(A) promoting a viable population increase
(B) supporting sustainable population explosion
(C) fostering an unbalanced population growth
(D) discouraging a sustainable population upswing

Q.4

Select the most appropriate adjective to fill in the blank. They needed a _____ answer for every question.

(A) most precise
(B) preciser
(C) precise
(D) more precise

Q.5

Select the most appropriate ANTONYM of the given word. Haste

(A) Delay
(B) Quick
(C) Twist
(D) Speed

Q.6

Select the most appropriate option that can substitute the underlined words in the given sentence. The walls of the marriage hall were “covered by” beautiful purple curtains.

(A) covered of
(B) covered with
(C) covered on
(D) covered upon

Q.7

Based on the situation in the given sentence, select the most appropriate idiom for the underlined segment. Having known to the facts, the manager “disregarded the arguments of” the customer.

(A) Bite the Bullet
(B) Turned out
(C) Turned a deaf ear
(D) Turned against

Q.8

Select the most appropriate synonym of the word 'Pamper' from the given sentence. I always feel that nuclear families always mollycoddle their kids without directing them to the correct path or denying their unrealistic demands.

(A) Demands
(B) Mollycoddle
(C) Nuclear
(D) Path

Q.9

Select the sentence that contains a spelling error.

(A) He has made significant improvement in the last few weeks.
(B) Despite his formidable appearance, he possesses a kind heart.
(C) They had been disgrantled with the company due to various reasons.
(D) We saw a pride of lions nonchalantly roaming in the forest.

Q.10

Select the most appropriate synonym of the given word. Vigorous

(A) Guarded
(B) Furious
(C) Energetic
(D) Academic

Q.11

Select the correct spelling of the underlined word. She has a “magnificient” personality.

(A) Magnifecient
(B) Magnificent
(C) Megnifecent
(D) Megnificent

Q.12

Select the most appropriate ANTONYM of the word given in brackets to fill in the blank. He was able to do _____(tedious) jobs and hence decided to be in the airline service, which demanded 12 long hours of service.

(A) timid
(B) riotous
(C) irksome
(D) delightful

Q.13

Change the following sentence into positive degree. Yash is not better than Harsh.

(A) Yash is as good as Harsh.
(B) Yash is not good than Harsh.
(C) Yash and Harsh both are good.
(D) Yash is best than Harsh.

Q.14

Select the option that rectifies the spelling of the underlined word in the given sentence. The children were excited to ride the “carusel.”

(A) Carrousel
(B) Karusel
(C) Carousel
(D) Carousell

Q.15

Select the most appropriate option to fill in the blanks. If 2023 was ___ year consumers helped stave off a recession with an unexpected surge in spending, many economists _____ 2024 will be the time shoppers finally tighten their belts.

(A) a, wanted
(B) the, predict
(C) a, see
(D) an, thought

Q.16

In the following sentence the underlined part contains an error. Alternatives to the underlined part are given as options. Select the correct alternative. Playing ‘hide-n-seek’ is the one of the “fond” memories of my childhood.

(A) fonder
(B) fondest
(C) most fondest
(D) more fonder

Q.17

The given sentence is divided into four segments. Select the option that has the segment with a grammatical error. I left my / coat in the / hospital when I / had visit Alex.

(A) I left my
(B) had visit Alex.
(C) coat in the
(D) hospital when I

Q.18

Select the most appropriate option to fill in the blank. The old man shared his _____ wisdom with the young generation.

(A) frivolous
(B) ephemeral
(C) trivial
(D) sagacious

Q.19

Select the most appropriate ANTONYM of the given word. Contrary

(A) Reverse
(B) Transformational
(C) Similar
(D) Forward

Q.20

Select the most appropriate meaning of the given idiom. Clear as mud

(A) Very easy to understand
(B) A dirty surrounding
(C) Very difficult to understand
(D) A clean surface

Q.21

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. We should remember that all religions (1)_____ good conduct in life, the foundation of which should rest on (2)_____. Without the basis of morality, philosophy would remain mere polemics, and religion would be (3)_____ and vain. Nobody can say that he is religious unless he is (4)_____ and unselfish. It is realised that by constant endeavour through thought, word and deed, one could (5)_____ absolute peace of mind.

Select the most appropriate option to fill in blank number 1.

(A) preach
(B) depart
(C) refrain
(D) abandon

Q.22

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. We should remember that all religions (1)_____ good conduct in life, the foundation of which should rest on (2)_____. Without the basis of morality, philosophy would remain mere polemics, and religion would be (3)_____ and vain. Nobody can say that he is religious unless he is (4)_____ and unselfish. It is realised that by constant endeavour through thought, word and deed, one could (5)_____ absolute peace of mind.

Select the most appropriate option to fill in blank number 2.

(A) unconscientiousness
(B) evil
(C) ethics
(D) boundaries

Q.23

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. We should remember that all religions (1)_____ good conduct in life, the foundation of which should rest on (2)_____. Without the basis of morality, philosophy would remain mere polemics, and religion would be (3)_____ and vain. Nobody can say that he is religious unless he is (4)_____ and unselfish. It is realised that by constant endeavour through thought, word and deed, one could (5)_____ absolute peace of mind.

Select the most appropriate option to fill in blank number 3.

(A) useful
(B) popular
(C) meaningless
(D) rational

Q.24

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. We should remember that all religions (1)_____ good conduct in life, the foundation of which should rest on (2)_____. Without the basis of morality, philosophy would remain mere polemics, and religion would be (3)_____ and vain. Nobody can say that he is religious unless he is (4)_____ and unselfish. It is realised that by constant endeavour through thought, word and deed, one could (5)_____ absolute peace of mind.

Select the most appropriate option to fill in blank number 4.

(A) grotesque
(B) truthful
(C) lethargic
(D) insensitive

Q.25

In the following passage, some words have been deleted. Read the passage carefully and select the most appropriate option to fill in each blank. We should remember that all religions (1)_____ good conduct in life, the foundation of which should rest on (2)_____. Without the basis of morality, philosophy would remain mere polemics, and religion would be (3)_____ and vain. Nobody can say that he is religious unless he is (4)_____ and unselfish. It is realised that by constant endeavour through thought, word and deed, one could (5)_____ absolute peace of mind.

Select the most appropriate option to fill in blank number 5.

(A) scatter

(B) forfeit

(C) divide

(D) acquire

Answer key

Numerical and Mathematical Ability

Q.1 **(D)**	Q.2 **(A)**	Q.3 **(C)**	Q.6 **(B)**	Q.7 **(D)**	Q.8 **(A)**
Q.10 **(B)**	Q.11 **(B)**	Q.12 **(A)**	Q.13 **(C)**	Q.14 **(C)**	Q.15 **(C)**
Q.16 **(D)**	Q.17 **(A)**	Q.18 **(A)**			

Reasoning Ability and Problem Solving

Q.1 **(B)**	Q.2 **(C)**	Q.5 **(C)**	Q.8 **(B)**	Q.9 **(B)**	Q.10 **(A)**
Q.11 **(D)**	Q.12 **(B)**	Q.13 **(C)**	Q.14 **(C)**	Q.16 **(A)**	Q.17 **(B)**
Q.18 **(B)**	Q.19 **(B)**	Q.20 **(D)**			

General Awareness

Q.1 **(D)**	Q.2 **(B)**	Q.3 **(D)**	Q.4 **(B)**	Q.5 **(C)**	Q.6 **(A)**
Q.7 **(C)**	Q.8 **(D)**	Q.9 **(B)**	Q.10 **(D)**	Q.11 **(A)**	Q.12 **(B)**
Q.13 **(A)**	Q.14 **(A)**	Q.15 **(D)**	Q.16 **(C)**	Q.17 **(D)**	Q.18 **(B)**
Q.19 **(B)**	Q.21 **(D)**	Q.22 **(C)**	Q.23 **(B)**	Q.24 **(B)**	Q.25 **(A)**

English Language and Comprehension

Q.1 **(C)**	Q.2 **(A)**	Q.3 **(A)**	Q.4 **(C)**	Q.5 **(A)**	Q.6 **(B)**
Q.7 **(C)**	Q.8 **(B)**	Q.9 **(C)**	Q.10 **(C)**	Q.11 **(B)**	Q.12 **(D)**
Q.13 **(A)**	Q.14 **(C)**	Q.15 **(B)**	Q.16 **(B)**	Q.17 **(B)**	Q.18 **(D)**
Q.19 **(C)**	Q.20 **(C)**	Q.21 **(A)**	Q.22 **(C)**	Q.23 **(C)**	Q.24 **(B)**
Q.25 **(D)**					

Explanations

Numerical and Mathematical Ability

Q.1 · Answer (D) 49.6

(40 + 49 + 56 + 65 + 38)/5 = 248/5 = 49.6 kg.

Q.2 · Answer (A) 6 : 4 : 5

The 360 toffees are first shared 80, 120 and 160. Pushkal then gets 20% of 120 = 24 and 25% of 160 = 40, so the shares become 144, 96 and 120 = 6 : 4 : 5.

Q.3 · Answer (C) ₹2,400

0.88C + 540 = 1.15C, so 0.27C = ₹540 and the cost price C = ₹2,000. For a 20% gain: 1.2 × 2,000 = ₹2,400.

Q.6 · Answer (B) 32 km/h, 24 km/h

With the fast train at f km/h, 96/(f – 8) – 96/f = 1, so $f^2 – 8f – 768 = 0$ and f = 32. The slow train runs at 24 km/h: 96/24 – 96/32 = 4 – 3 = 1 hour.

Q.7 · Answer (D) 22 km/h

The car takes 80/20 + 30/30 = 5 hours for 110 km: 110/5 = 22 km/h.

Q.8 · Answer (A) 139

The class weighed 40 × 60 = 2,400 kg; the 38 left weigh 38 × 59.5 = 2,261 kg. The two who left weighed 2,400 – 2,261 = 139 kg.

Q.10 · Answer (B) 49.8

(40 + 49 + 56 + 66 + 38)/5 = 249/5 = 49.8 kg.

Q.11 · Answer (B) 720

SI = 4,000 × 3 × 6/100 = ₹720.

Q.12 · Answer (A) 15 cm²

The lines from the centroid to the vertices divide a triangle into three triangles of equal area, so ΔGYZ = 45/3 = 15 cm^2.

Q.13 · Answer (C) 6,240

The 22% discount is the ₹1,760 saved, so the marked price is 1,760/0.22 = ₹8,000 and he paid 78% of it: ₹6,240.

Q.14 · Answer (C) 25%

The profit on 20 cakes is the selling price of 4, so their cost is the selling price of 16. Profit = 4/16 × 100 = 25%.

Q.15 · Answer (C) 31.82%

Annual income = 55,000 × 12 = ₹6,60,000, so savings = ₹2,10,000. 2,10,000/6,60,000 × 100 = 31.82%.

Q.16 · Answer (D) 81.5

(3 × 83 + 4 × 76 + 5 × 85)/12 = (249 + 304 + 425)/12 = 978/12 = 81.5.

Q.17 · Answer (A) 750

P = SI × 100/(R × T) = 240 × 100/(16 × 2) = ₹750.

Q.18 · Answer (A) 27

513 = 3^3 × 19, 1134 = 2 × 3^4 × 7 and 1215 = 3^5 × 5. The highest common factor is 3^3 = 27.

Reasoning Ability and Problem Solving

Q.1 · Answer (B) Only conclusion [1] follows

Some pencils are ACs, and every pencil is a remote, so those ACs are remotes: [1] follows. Nothing says every remote is an AC, so [2] does not.

Q.2 · Answer (C) × and +

Interchanging × and +: 15 + 14 – 8 × 12 ÷ 4 = 15 + 14 – 24 = 5.

Q.5 · Answer (C) Only conclusion II follows
'All leaves are trees' is given, so II follows. The trees that are plants need not include any leaves, so I does not follow.

Q.8 · Answer (B) M
Facing the centre, going round to the right of L: L, M, A, B, C, X, Y. Y must be on L's left (on L's right, X would need L's seat), X to Y's left and B third to L's right, which leaves M then A in the two seats after L. Fourth to the right of C is M.

Q.9 · Answer (B) –269
After the interchange: 48 – 16 × 20 + 21 ÷ 7 = 48 – 320 + 3 = –269.

Q.10 · Answer (A) 119
Each letter takes its position counted from Z (Z = 1, A = 26) and the values are added: HOUSE = 19 + 12 + 6 + 8 + 22 = 67 and GROUND = 20 + 9 + 12 + 6 + 13 + 23 = 83. KITCHEN = 16 + 18 + 7 + 24 + 19 + 22 + 13 = 119.

Q.11 · Answer (D) RUVT
Each letter moves 6 places forward: N → T, Q → W, R → X, P → V, and FHIG → LNOM. So LOPN → RUVT.

Q.12 · Answer (B) -151
Each term is 31 less than the one before: –120 – 31 = –151.

Q.13 · Answer (C) 363
Each number is multiplied by 11: 16 × 11 = 176 and 21 × 11 = 231. So 33 × 11 = 363.

Q.14 · Answer (C) Q
Facing the centre, going round to the left of R: R, L, M, N, O, P, Q. With three seats between R and O on R's left, N next to O and only M between L and N, L, M and N must fill those three seats. P, third to L's right, is O's left neighbour, so Q is second to O's left.

Q.16 · Answer (A) 63 : 28
The product of the first number's digits plus 10 gives the second: 4 × 8 + 10 = 42. Likewise 6 × 3 + 10 = 28; the other pairs do not fit (48 would give 42, not 24; 69 gives 64 and 65 gives 40).

Q.17 · Answer (B) KPO
First letters move back 2 (S, Q, O, M, K), second letters back 2 (X, V, T, R, P) and third letters forward 2 (G, I, K, M, O): KPO.

Q.18 · Answer (B) 2
P is the only letter in both words and 2 the only digit in both codes, so P = 2.

Q.19 · Answer (B) UPXT
Each letter moves 2 places forward: MHPL → OJRN and PKSO → RMUQ. So SNVR → UPXT.

Q.20 · Answer (D) 5
MASTER and STREAK share S, T, R, E and A, and their codes share 9, 1, 7, 3 and 2. The letter left in STREAK is K and the digit left in its code is 5, so K = 5.

General Awareness

Q.1 · Answer (D) Bismillah Khan
'Sange Meel Se Mulaqat', made by Goutam Ghose, is a documentary on the shehnai maestro Ustad Bismillah Khan.

Q.2 · Answer (B) West Bengal
Kalbaisakhi, 'the calamity of the month of Baisakh', is West Bengal's name for the violent evening thunderstorms (nor'westers) of the hot season.

Q.3 · Answer (D) Starch, oils and protein granules are stored in chromoplasts.
Starch, oil and protein granules are stored in leucoplasts, the colourless plastids; chromoplasts hold pigments. Plastids do have their own DNA and ribosomes, and the chloroplast's matrix is the stroma.

Q.4 · Answer (B) Mehrauli iron pillar
The Mehrauli iron pillar dates from the Gupta period (about AD 400), long after India's Iron Age. Iron smelting, Painted Grey Ware and Northern Black Polished Ware all belong to the Iron Age.

Q.5 · Answer (C) Financial Emergency
Article 360 provides for a Financial Emergency; a National Emergency is declared under Article 352 and President's Rule in a state under Article 356.

Q.6 · Answer (A) All a, b and c
SSC's key gives all three. The Union Budget sets out the government's priorities and details each ministry's expenditure; b is looser, as the Budget is chiefly a statement of estimates for the coming year, though it also reports the previous year's actual receipts and spending.

Q.7 · Answer (C) Ghiyasuddin Balban
Ghiyasuddin Balban chose to consolidate the Delhi Sultanate rather than expand it, crushing rebels and enemies with his policy of 'blood and iron'.

Q.8 · Answer (D) Bricks
The stupa at Piprahwa, where a casket of Buddhist relics was found, is built of bricks.

Q.9 · Answer (B) Hunter Committee
The Hunter Committee (the Disorders Inquiry Committee) was set up in 1919 to inquire into the Jallianwala Bagh massacre.

Q.10 · Answer (D) Annelida
SSC's key gives Annelida, the phylum of ring-segmented worms such as the earthworm, whose bodies show metameric segmentation. Arthropods have segmented bodies too, so Arthropoda is also arguable; molluscs are not segmented.

Q.11 · Answer (A) 10th
The 10th season of the Pro Kabaddi League began on 2 December 2023.

Q.12 · Answer (B) Rengma
Ngada is the harvest festival of the Rengma Nagas, and the Rengma war dance is performed during it. Jawara is from Madhya Pradesh, Dalkhai from Odisha and Garadi from Puducherry.

Q.13 · Answer (A) Gambit
A gambit is an opening in which a player sacrifices material, usually a pawn, for compensation such as faster development or the initiative. Castling is a move of the king and a rook.

Q.14 · Answer (A) 8 m/s for both
Without air resistance every body falls with the same acceleration, whatever its mass: $v = \sqrt{(2gh)} = \sqrt{(2 \times 10 \times 3.2)} = \sqrt{64} = 8$ m/s for both.

Q.15 · Answer (D) Atmanirbhar Bharat Abhiyan
The Atmanirbhar Bharat Abhiyan (self-reliant India campaign) was announced in May 2020.

Q.16 · Answer (C) Jhumpa Lahiri
'The Namesake' and 'The Lowland' are Jhumpa Lahiri's novels, and 'Unaccustomed Earth' is her collection of short stories.

Q.17 · Answer (D) Egypt
Egypt conferred the Order of the Nile, its highest state honour, on Narendra Modi during his visit to Cairo in June 2023.

Q.18 · Answer (B) Five
The New India Literacy Programme has five components: foundational literacy and numeracy, critical life skills, basic education, vocational skills development and continuing education.

Q.19 · Answer (B) Australia
The Concurrent List was borrowed from the Australian Constitution, as was the joint sitting of the two Houses of Parliament.

Q.21 · Answer (D) 5.27%
NITI Aayog's National Multidimensional Poverty Index 2023 puts urban India's headcount ratio at 5.27% in 2019-21, down from 8.65% in 2015-16.

Q.22 · Answer (C) 31 October
National Unity Day (Rashtriya Ekta Diwas) is observed on 31 October, the birth anniversary of Sardar Vallabhbhai Patel.

Q.23 · Answer (B) rock cycle
The rock cycle is the continuous process by which igneous, sedimentary and metamorphic rocks change into one another.

Q.24 · Answer (B) Dance
Mallika Sarabhai, a Kuchipudi and Bharatanatyam dancer, received the Padma Bhushan in 2010 for her work in dance.

Q.25 · Answer (A) 2000
India's National Population Policy was released in 2000, with the long-term aim of a stable population by 2045.

English Language and Comprehension

Q.1 · Answer (C) The teacher from the South explained the theory in a detailed manner.
A region takes the definite article, 'the South', and 'in a detailed manner' is the set phrase; 'an South' and 'the detailed manners' are wrong.

Q.2 · Answer (A) break into
To 'break into' a market is to enter it successfully; 'break down', 'break off' and 'break out' mean collapse, stop and erupt.

Q.3 · Answer (A) promoting a viable population increase
'Promoting a viable population increase' keeps the meaning: 'viable' matches 'sustainable' and 'increase' matches 'growth'. 'Explosion', 'unbalanced' and 'discouraging' change the sense.

Q.4 · Answer (C) precise
Nothing is being compared, so the positive degree 'precise' fits after 'a'; 'preciser' is not standard English.

Q.5 · Answer (A) Delay
'Haste' is hurry; its opposite is 'delay'.

Q.6 · Answer (B) covered with
A surface is 'covered with' the material that covers it: 'covered with beautiful purple curtains'.

Q.7 · Answer (C) Turned a deaf ear
To 'turn a deaf ear' to something is to refuse to listen to it, which is what the manager did.

Q.8 · Answer (B) Mollycoddle
To 'mollycoddle' is to pamper or overindulge.

Q.9 · Answer (C) They had been disgrantled with the company due to various reasons.
The correct spelling is 'disgruntled'.

Q.10 · Answer (C) Energetic
'Vigorous' means strong and active: 'energetic'.

Q.11 · Answer (B) Magnificent
The correct spelling is 'magnificent'.

Q.12 · Answer (D) delightful
'Tedious' means dull and tiresome; its opposite is 'delightful'. 'Irksome' is close to a synonym.

Q.13 · Answer (A) Yash is as good as Harsh.
SSC's key gives 'Yash is as good as Harsh', the usual positive-degree form of 'not better than' (strictly, 'not better' also allows 'worse'). 'Not good than' and 'best than' are ungrammatical, and 'both are good' drops the comparison.

Q.14 · Answer (C) Carousel
The standard spelling is 'carousel' ('carrousel' is a less common variant).

Q.15 · Answer (B) the, predict
'The year' points to a particular year, defined by the clause that follows, and economists 'predict' what 2024 will bring.

Q.16 · Answer (B) fondest
'One of the' is followed by a superlative: 'one of the fondest memories'. 'Most fondest' is a double superlative.

Q.17 · Answer (B) had visit Alex.
'Had' needs the past participle, not the base form: 'when I visited Alex' or 'had visited', not 'had visit'.

Q.18 · Answer (D) sagacious
'Sagacious' means wise and shrewd, which suits an old man's wisdom; 'frivolous', 'ephemeral' and 'trivial' do not.

Q.19 · Answer (C) Similar
'Contrary' means opposite in nature; its opposite is 'similar'.

Q.20 · Answer (C) Very difficult to understand
'Clear as mud' is said ironically of something not clear at all: very difficult to understand.

Q.21 · Answer (A) preach
All religions 'preach', that is teach and urge, good conduct in life.

Q.22 · Answer (C) ethics
Good conduct should rest on 'ethics', moral principles, which the next sentence calls 'the basis of morality'.

Q.23 · Answer (C) meaningless
Without morality religion would be 'meaningless' and vain, as philosophy would be mere polemics.

Q.24 · Answer (B) truthful
Being religious goes with being 'truthful' and unselfish.

Q.25 · Answer (D) acquire
Constant endeavour is how one gains something, so one could 'acquire' absolute peace of mind; 'scatter', 'forfeit' and 'divide' go against the sense.

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